

Dictionary :-

- 1) Dictionary is also one of the container data type in python
- 2) Dictionary is a collection of key-value pair.

Key-value pairs are separated by comma & enclosed in curly

brackets.

key : value Pair

eg. Student = { "name": "Gurush",
 1 ↗ ↖
 "age": 25,
 2 ↗ ↖
 "address": "mumbai"
 }

- 3) different keys may have same value.

eg. dict2 = { "name" : "janeesh",
"roll-no" : 25,

```
"age" : 25  
}
```

4) Keys should be unique but if we have some key with different value, it will consider latest key-value pair.

```
eg. dict3 = { "name": "ganesh",  
              "roll-no": 25,  
              "name": "mahesh"  
            }
```

```
print(dict3)
```

```
o/p - { "name": "mahesh",  
        "roll-no": 25 }
```

* Accessing dictionary elements -

i) To print/return the entire dictionary

we use dictionary name.

eg. dict1 = { "name" : "gareesh",
 "age" : 25 }

print(dict1) → { "name" : "gareesh",
 "age" : 25 }

print(type(dict1)) → <dict>

2) Using keys we can access/print the values.

print(dict1["name"]) → "gareesh"

print(dict1["age"]) → 25

* Indexing & Slicing does not work in dictionaries.

* Looping in Dictionaries -

If we want to process each

element in the dictionary, we

need to iterate the dictionary,
& we do it using for loop.

1) for loop - `items()` - ^{to iterate} pairs

eg. dict4 = { "name": "gonesh",
 "age": 25 }

```
for key, value in dict4.items():  
    print(key, value)
```

```
o/p - name gonesh  
      age 25
```

2) for loop - `keys()`
 To iterate keys

eg. dict5 = { "car": "Tata",
 "model": "nexon" }

```
for key in dict5.keys():  
    print(key)
```

o/p - car
model

3) for loop - values()

eg. dict6 = { "TV": "lg",
 "display": "OLED" }

```
for value in dict6.values():  
    print(value)
```

o/p- lg
OLED

4) enumerate :-

```
dict7 = { "mobile" : "iphone",  
          "model" : "iphone18" }
```

```
for i, (k, v) in enumerate(dict7.items()):  
    print(i, k, v)
```

```
0p - 0 mobile iphone  
    1 model iphone18
```

* Basic Dictionary Operations

String, tuple - immutable

List, set, dictionary - Mutable

int, float, complex, bool - immutable

1] Mutability - Dictionaries are mutable

once we create dictionary, we can change it later.

change → add, delete, replace

eg. dict1 = { "laptop": "Dell",
 "processor": "i5" }

dict1["Ram"] = "8Gb"

print(dict1)

o/p - { "laptop": "Dell",
 "processor": "i5",
 "Ram": "8Gb" }